Lösungen · Trigonometrie

1. $\mathbf{y = 160 \cdot \frac{\sin 141°}{\sin 34°} \approx 180{,}1\ cm}$ ..... $\frac{y}{\sin 141°} = \frac{160}{\sin 34°}$

2. $\mathbf{BC = 384 \cdot \frac{\sin 38°}{\sin 34°} \approx 422{,}8\ m}$ ..... BC : sin 38° = $\frac{384}{\sin 34°}$

3. **AD, $\delta = 21°$ und rechter Winkel bei E**
A ≈ 2,81 cm² ..... AE = 4,1 · sin 21°; DE = 4,1 · cos 21°

4. $\mathbf{AC = 12{,}0 \cdot \frac{\sin 55°}{\sin 5°} \approx 112{,}8\ m}$ ..... $\gamma = 35° - 30°$; AC : sin 55° = $\frac{12{,}0}{\sin 5°}$

5. $\mathbf{BD = 4{,}1 \cdot \frac{\sin 123°}{\sin 36°} \approx 5{,}85\ cm}$ ..... BD : sin 123° = $\frac{4{,}1}{\sin 36°}$

6. $\mathbf{BC = 4{,}50 \cdot \frac{\sin 62{,}8°}{\sin 34{,}9°} \approx 7{,}00\ m;\ A \approx 92{,}0\ m^2}$ ...
BC : sin 62,8° = $\frac{4{,}50}{\sin 34{,}9°}$; A = 8,00 · (4,50 + 7,00)

7. **Weg ≈ 5665 m** ..... JR : sin 60° = $\frac{2800}{\sin 50°}$; Weg = JR + 2500

8. **nein** ..... ∠DBC = 109° − 35° = 74°; BC : sin 60° = $\frac{2004}{\sin 74°}$; BC = 2004 · $\frac{\sin 60°}{\sin 74°}$ ≈ 1805,5 m

9. $\mathbf{DP = 11 \cdot \frac{\sin 115°}{\sin 49°} - 10 \approx 3{,}2\ m}$ ..... DE : sin 115° = $\frac{11}{\sin 49°}$; DP = DE − 10

10. $\mathbf{AD = 9 \cdot \frac{\sin 60°}{\sin 80°} \approx 7{,}9\ m}$ ..... ∠ACD = 86° − 26° = 60°; AD : sin 60° = $\frac{9}{\sin 80°}$